Logarithmic Functions & Properties

Converting between exponential and logarithmic form

1 Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.

2 Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.

Properties of logarithms

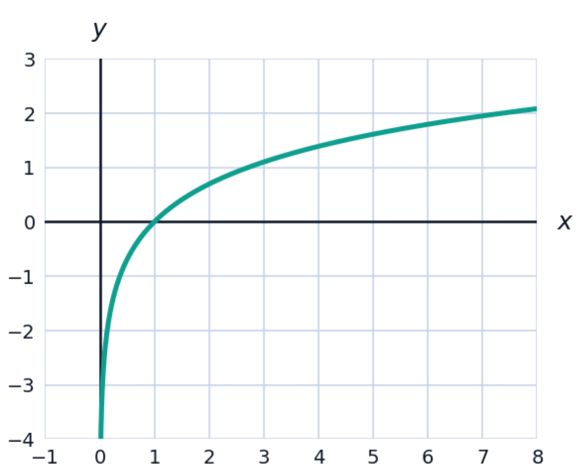
3 Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.

4 Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.

5 Use the change of base formula to evaluate $\log_7 50$, to three decimal places.

Visualizing logarithmic functions

6 The graph shows $f(x) = \ln(x)$, the natural logarithm.



a State the domain and the vertical asymptote shown in the graph.

b State the x-intercept shown.

Solving exponential equations using logarithms

7 Solve $5^x = 40$ for x, to three decimal places.

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8 Solve $3(2)^{x-1} = 45$ for x, to three decimal places.

Solving logarithmic equations

9 Solve $\log_2(x+3) = 4$ for x, and check that the solution keeps the argument positive.

10 Solve $\log(x) + \log(x-3) = 1$ for x, checking both solutions against the domain.

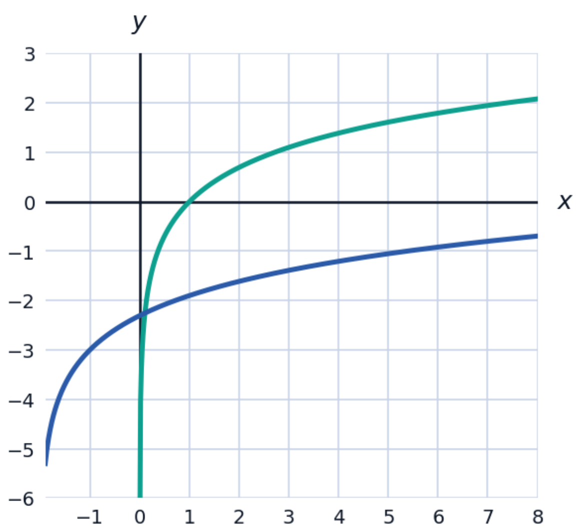
Applications: the pH scale and decibels

11 The pH of a solution is given by pH $= -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.

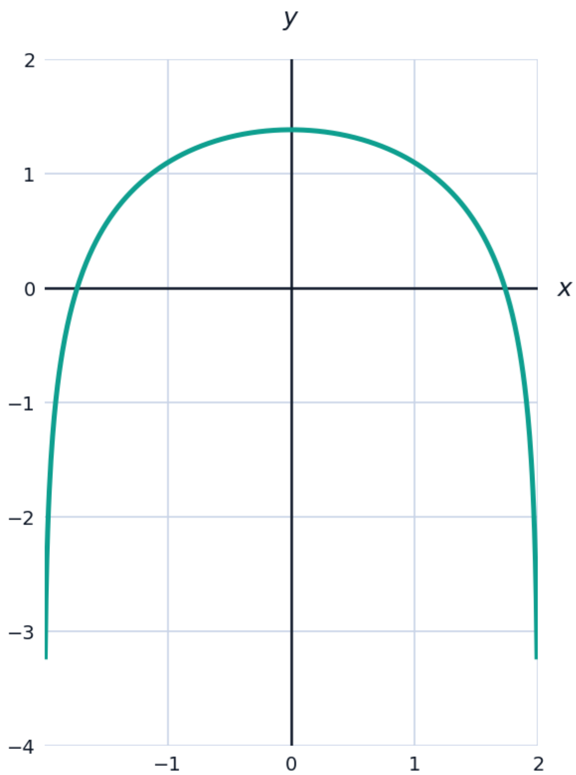
12 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.

Transformations of logarithmic functions

13 Describe how the graph of $g(x) = \ln(x+2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.



14 Find the domain of $h(x) = \log(4 - x^2)$.



Solving equations that combine exponentials and logarithms

15 Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.

16 Solve $\log_2(x) + \log_2(x - 2) = 3$ for x, checking the domain.

Comparing logarithmic growth to other function families

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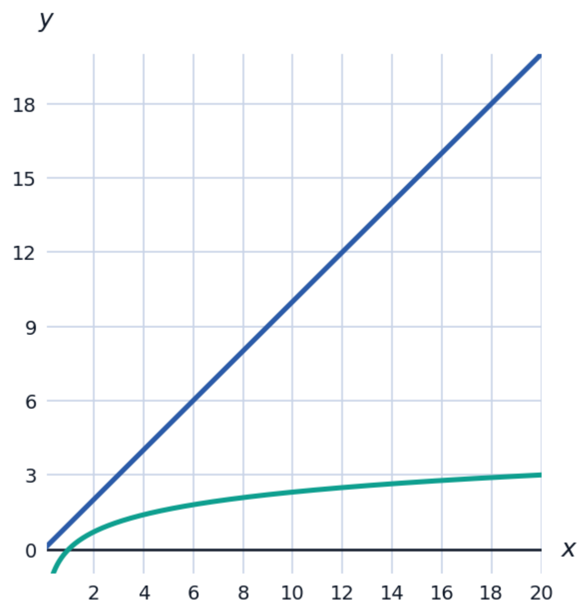
17 Explain why logarithmic functions grow much more slowly than linear functions for large x, and give a numerical comparison of $\log(x)$ and x at $x = 1000$.



Modeling with logarithms: the Richter scale

18 Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

19 If an earthquake's intensity is $10{,}000$ times a reference intensity I_0, find its Richter magnitude.